

MAHESH YADDANAPUDI

Software Architect & Platform Engineer | Agentic AI Platforms, Workflow & Orchestration, Distributed Systems

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SUMMARY

Software architect and platform engineer with 17+ years designing distributed backend systems, workflow and orchestration engines, and agentic AI platforms at Workday, Oracle (OCI), Google Cloud, JPMorgan Chase and Verizon. At Workday I built the MCP host that became the architectural baseline for the product's official MCP host, and I initiated and led the design of APFlow, a reusable workflow engine. I've led engineering teams, run build-vs-adopt evaluations, and built multi-agent platforms end to end in Java, Python and TypeScript.

CORE SKILLS

- **Agentic AI & LLM:** Model Context Protocol (hosts, servers, tool gateways), A2A, multi-agent orchestration (LangGraph, Spring AI), tool/skill/agent registries, CodeAct, human-in-the-loop approvals, guardrails and policy enforcement, LLM-as-judge evaluation, prompt engineering, LoRA/QLoRA fine-tuning (MLX)
- **Orchestration & workflow:** Workflow/BPMN engines (Flowable, Netflix Conductor), process lifecycle and extension-model design, event-driven state management (Kafka), schedulers, checkpoint/resume, DAG workflows, infrastructure automation (Ansible)
- **Reliability & observability:** Production incident stabilization, crash recovery, readiness-first deploys, leader election, backup/restore, rate limiting and admission control, OpenTelemetry, LangSmith, Prometheus/Grafana, Snyk/Semgrep remediation, JDK migrations
- **Data & retrieval:** PostgreSQL/pgvector, Kafka, Redis, Elasticsearch, Neo4j, BM25, hybrid retrieval and GraphRAG (reciprocal rank fusion), embeddings, ML forecasting pipelines (N-BEATS, Prophet)
- **Languages & infra:** Java, Python, TypeScript; Spring Boot, FastAPI, React, Angular; Docker, Kubernetes, CI/CD (Jenkins); GCP, AWS, Oracle Cloud (OCI); microservices, REST, SOA

EXPERIENCE

Senior Machine Learning Engineer, Workday (Adaptive Planning) | Pleasanton / Santa Clara, CA Sep 2022 - Present

Senior Machine Learning Engineer, Feb 2026 - Present; Software Engineer, Sep 2022 - Feb 2026 (promoted)

- Built the MCP Host proof of concept (Java, Spring AI) that became the architectural baseline for the official MCP planning host, which another team adopted and now owns. It shaped the MCP-based assistant architecture for the Planning product.
- Initiated and led the design of APFlow, a generic, pluggable process/workflow engine behind Adaptive Planning Configurable Workflows (shipped 2026R1, demoed at Workday Rising). The product feature closed a workflow gap Gartner flagged in its Dec 2022 Magic Quadrant.
- Defined APFlow's core interfaces, lifecycle operations and extension model so other teams (dashboarding, planning, discovery) can reuse it. Ran the build-vs-adopt evaluation of Flowable and Netflix Conductor and presented the design to the architects council.
- Contributed to Workday Adaptive Planning's shipped agent.
- Built an MCP server proof of concept that exposes Configurable Workflow APIs as agent tools. Now leading MCP-integrated workflow monitoring end to end (assistant > MCP host > MCP server > Planning APIs) and Hierarchy Submit.
- Built event-driven workflow state management on Kafka for APFlow. Integrated Flowable into Planning for Shared Scenarios, building the Send Back approval flow (record locking, notifications) and workflow-instance update logic for long-running workflow state.
- Owned the backend for Predictive Forecaster, Adaptive Planning's ML forecasting product: scheduler integration, create/update/pause/resume of scheduled forecasts, timezone handling and OpenAPI documentation, with dispatch to an ML pipeline (N-BEATS, Prophet).
- Production reliability: stabilized a critical ML pipeline outage, resolved anomaly-detection hangs, migrated the AI/ML service from JDK 8 to 17, remediated Snyk/Semgrep findings and led the spike to retire legacy ML infrastructure.
- Drove AI-first engineering adoption across the team (Claude, Cursor). AI-assisted design reached 7 APFlow design iterations in one week, saving an estimated full team release of effort.

INDEPENDENT PROJECTS (PUBLIC UNLESS MARKED PRIVATE)

Concierge Agent | Registry-driven multi-agent platform (private repo; source available on request) | Python, FastAPI, LangGraph, PostgreSQL/pgvector, React, Docker

- Tools > Skills > Sub-agents registries in PostgreSQL. Skills are declarative markdown manifests bound strictly to their tools. MCP servers plug in at runtime and their tools are callable on the next loop iteration. Sub-agents are validated DAGs (branching, parallel fan-out, error edges, human-in-the-loop approval) compiled into LangGraph subgraphs. Supports two orchestrator modes and outbound A2A delegation.

- Safety and bounded work: run wall-clocks, call limits, admission control with queueing and load shedding, a named kill switch for every autonomous behavior, a single fencing choke point for untrusted input, and an egress allow-list policy.
- Recovery and operations: PostgreSQL checkpoints, drain on shutdown and reaping on restart, stream resume across deploys, readiness-first deploys, a multi-replica control plane with leader election and cross-replica cancellation, and backup/restore with a measured RTO.
- Tracing and evals: always-on run traces plus OpenTelemetry and LangSmith; dataset-driven evals (exact, contains and LLM-as-judge graders) on the production run path; a provider-agnostic model port (Anthropic, Gemini, OpenAI, OpenRouter). 1,068 backend tests passing.

J.A.R.V.I.S | *Local-first agent OS* | TypeScript, Python, Next.js, Rust (Tauri), PostgreSQL/pgvector, OpenTelemetry

- A trust core routes every action through one gated loop: deny-first policy engine > approval broker > execution > independent verification > hash-chained audit, with an emergency stop in every interface. Includes an MCP host with trust-gated tools, a provider-agnostic model gateway with role-based routing, a local secrets vault, AES-256-GCM encrypted memory and a whole-stack acceptance harness.

MAOS | *Config-driven multi-agent orchestration* | Python, LangGraph, FastAPI, MCP, A2A

- A LangGraph supervisor routes work to workers on a framework-agnostic agent runtime (7+ agent frameworks, chosen per agent in YAML), with MCP tools over stdio/HTTP, A2A agent-card discovery, human-in-the-loop interrupt/resume backed by a PostgreSQL checkpointer, and an OpenAI-compatible streaming API.

MY-Manus | *CodeAct agent platform* | Java 17, Spring Boot, Spring AI, React, Docker

- Agents write and run Python in resource-limited Docker sandboxes, with 22 built-in tools plus dynamic MCP tool discovery, session replay, Prometheus metrics, JWT auth and rate limiting.

Other public repos

- TestRail MCP Server (Spring AI): 103 operations behind 4 Lucene-backed meta-tools to avoid MCP tool bloat, with local-only credentials. Trilayer: hybrid pgvector + keyword + Neo4j graph retrieval fused with reciprocal rank fusion (LLM-as-judge evaluation planned). Netflix Conductor ecosystem: conductor-ng-ui (25 stars, 6 forks), conductor-boot, and a Spring Boot starter published to Maven Central. LoRA fine-tuned code and documentation models published on Hugging Face.

PRIOR EXPERIENCE

Principal Member of Technical Staff (PMTS), OCI Block Storage, Oracle | Seattle, WA Jan 2022 - Sep 2022

- Core backend PMTS for OCI Block Storage, the service behind block volumes and boot volumes for OCI compute.
- Designed and built control-plane and data-plane features, focused on reliability, durability and performance at cloud scale.
- Joined design reviews, code reviews and on-call for a tier-0 service, and improved operational tooling and monitoring.

Software Engineer, Google | Warsaw, Poland May 2021 - Dec 2021

- Provided full-stack expertise for Google Kubernetes Engine, Anthos and serverless products, including scalability issues involving large data volumes; built investigation tooling to diagnose and resolve customer issues on Google Cloud.
- Worked as a subject-matter expert for engineering, sales and customer stakeholders, partnering with SRE and product teams on quality, and contributed to Cloud Console products.

Software Engineering Lead, JPMorgan Chase & Co. | Hyderabad, India Dec 2016 - May 2021

- Engineering manager and architect for an org-wide, no-code automation suite for production management (incident, change and problem-management automation), built on Netflix Conductor workflow orchestration, Ansible, Spring Boot, Spring Security, REST APIs and Docker/Kubernetes.
- Led a team of Java, infrastructure and Ansible/Python developers plus a Java lead. Owned estimates, resourcing, SIT/UAT, production, DR and maintenance activities, and automated recurring operational tasks.
- Open-sourced related Conductor work: a Spring Boot wrapper, a Spring Boot starter and an Angular management UI (public on GitHub).

Also: Senior Consultant Project Lead, Virtusa, Hyderabad (Apr 2016 - Nov 2016): Project lead for client delivery. Managed team delivery, client relationships and release management, and coordinated cross-functional teams so projects shipped on schedule. System Analyst, ValueLabs, Malaysia (Jan 2014 - Apr 2016): led a WebSphere DataPower gateway migration and designed a Personal Data Protection portal. Analyst, System Development, Verizon Data Services, Hyderabad (Dec 2011 - Jan 2014): SOA order-management integration platform (MQ, BPEL, web services). Software Engineer and Assistant Systems Engineer, Tata Consultancy Services, Chennai (Feb 2009 - Dec 2011).

EDUCATION & CERTIFICATIONS

- Bachelor of Engineering, Electronics and Instrumentation Engineering, St. Joseph's College of Engineering, Chennai (2004 - 2008)
- Introduction to Generative AI for Software Development, DeepLearning.AI (2025); Spring Boot Microservices and Spring Cloud; Jenkins (CI/CD); Mastering Ansible (Udemy, 2020); IBM WebSphere Certified SOA Associate and Solutions Developer (4 IBM certifications)